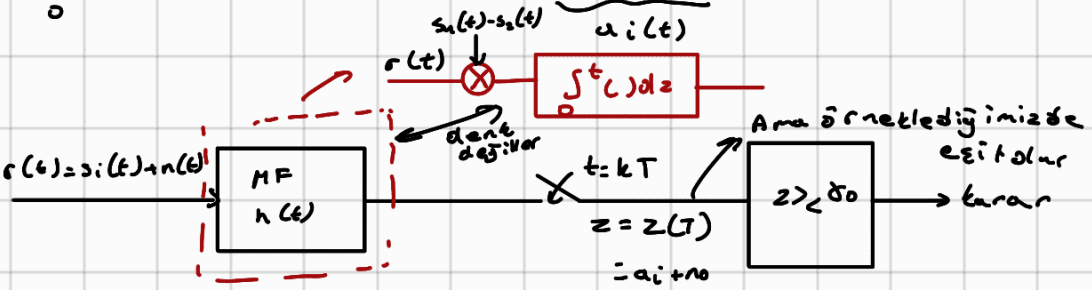
Genel

$$\gamma_0 = \frac{\sigma_0^2}{a_1 - a_2} \ln \frac{P(s_2)}{P(s_1)} + \frac{a_1 + a_2}{2} \rightarrow P_b = \left[1 - Q\left(\frac{\gamma_0 - a_1}{\sigma_0}\right)\right] P(s_1) + Q\left(\frac{\gamma_0 - a_2}{\sigma_0}\right) P(s_2)$$

$P(s_1) = P(s_2) = \frac{1}{2}$

$$\gamma_0 = \frac{a_1 + a_2}{2} \rightarrow P_b = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right) = Q\left(\sqrt{\frac{E_d}{2N_0}}\right) = Q\left(\sqrt{\frac{E_b}{N_0}(1-\rho)}\right)$$

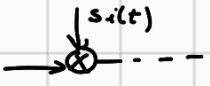
$$a_i = \int_0^T s_i(t)\left[s_1(t) - s_2(t)\right] dt = \underbrace{s_i(t) \circledast h(t)}_{a_i(t)}\Big|_{t=T}$$



$$h(t) = s_1(T-t) - s_2(T-t)$$

! Formüllerin nasıl çıktığını öğren

Soruda $h(t)$ $s_1(t)$'yi karşılıyor dersek $h(t) = s_1(T-t)$ olur

integral devresinde 

$$\sigma_0^2 = \frac{N_0}{2} E_h$$

$$E_d = \int_0^T \left[s_1(t) - s_2(t)\right]^2 dt \qquad E_b = \sum_{i=1}^{2} P(s_i) E_{si}$$

$$\rho = \frac{1}{E_b} \int_0^T s_1(t)\, s_2(t)\, dt$$

Örnek Unipolar signaling

"1" $\rightarrow s_1(t) = A,\ 0 \le t \le T$

"0" $\rightarrow s_2(t) = 0,\ 0 \le t \le T$

$P(s_1) = \frac{1}{4}$, $P(s_2) = \frac{3}{4}$

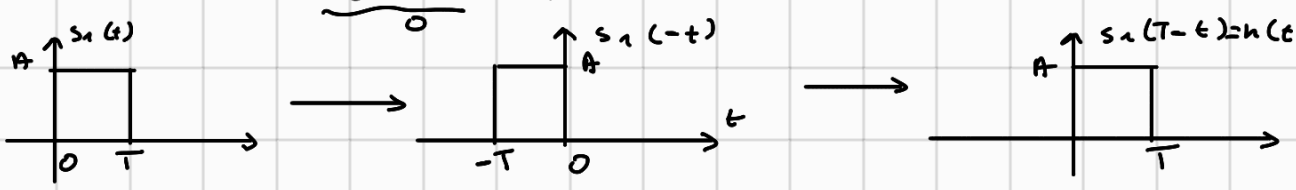
$A = 1\ V$, $T = 10^{-3}\ sn$, $\frac{N_0}{2} = 10^{-3}\ \frac{Watt}{Hz}$

$\gamma_0 = ?$ $P_b = ?$

$a_1 = A^2 T = 1^2 \cdot 10^{-3} = 10^{-3}$

$a_2 = 0$

$h(t) = s_1(T-t) - \underbrace{s_2(T-t)}_{0} = s_1(T-t)$



$$E_n = \int_0^T h^2(t)dt = \int_0^T A^2 dt = A^2 T = 10^{-3}$$

$$\sigma_0^2 = \frac{N_0}{2} \cdot E_n = 10^{-3} \cdot 10^{-3} = 10^{-6}$$

$$\gamma_0 = \frac{\sigma_0^2}{a_1 - a_2} \ln \frac{P(s_2)}{P(s_1)} + \frac{a_1 + a_2}{2}$$

$$= \frac{10^{-6}}{10^{-3}} \ln \frac{3/4}{1/4} + \frac{10^{-3} + 0}{2} = 1,5986 \cdot 10^{-3} \simeq 1,6 \cdot 10^{-3}$$

$$P_D = \left[1 - Q\left(\frac{\gamma_0 - a_1}{\sigma_0} \right) \right] P(s_1) + Q\left(\frac{\gamma_0 - a_2}{\sigma_0} \right) P(s_2)$$

($\gamma_0 \to 1,6 \cdot 10^{-3}$, $a_1 \to 10^{-3}$, $\sigma_0 \to 10^{-3}$, $P(s_1) \to \frac{1}{4}$; $\gamma_0 \to 1,6 \cdot 10^{-3}$, $a_2 \to 0$, $\sigma_0 \to 10^{-3}$, $P(s_2) \to \frac{3}{4}$)

$$P_D = \left[1 - \underbrace{Q(0,6)}_{2,7426 \cdot 10^{-1}} \right] \frac{1}{4} + \underbrace{Q(1,6)}_{5,4799 \cdot 10^{-2}} \cdot \frac{3}{4}$$

$$= 0,2225$$

Örnek

Transfer function of a MF is $H(f) = \frac{2 - 2e^{-j2\pi fT}}{j2\pi f}$

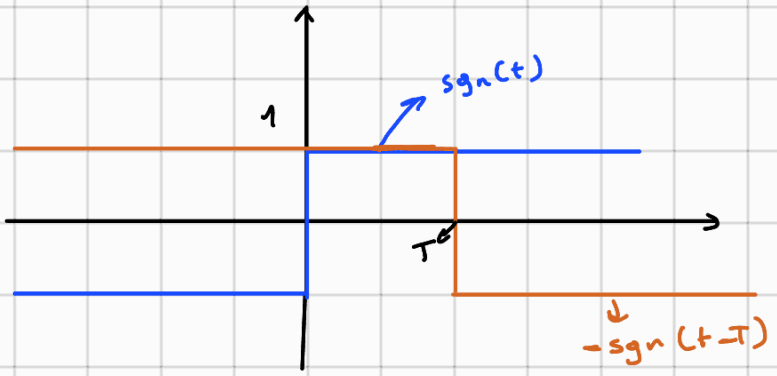
a) Find impulse response of MF and sketch it.

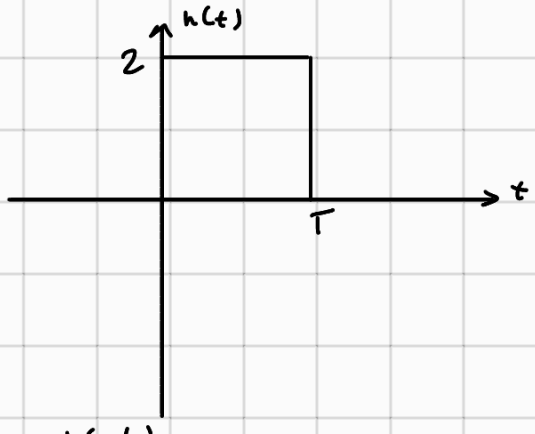
b) If the pulse $s(t)$ matched by this MF an sketch it.

c) If this $s(t)$ is applied to the input of MF, find the output signal of the MF and sketch it. Find the value at the sample instant.

d) If this $s(t)$ is applied to the input of the corelator, find the output signal of the correlator and sketch it. Find the value at the simple instant.

a) $H(f) = \frac{2}{j2\pi f} - \frac{2e^{-j2\pi fT}}{j2\pi f} \xrightarrow{\mathcal{F}^{-1}} h(t) = sgn(t) - sgn(t-T)$

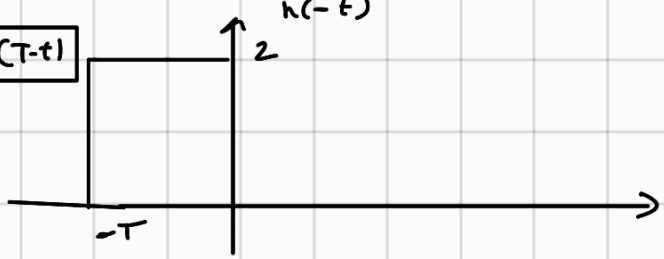


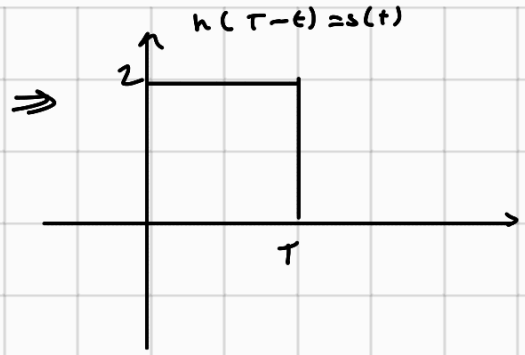


b) $h(t) = s(\underbrace{T-t}_{t'}) \longrightarrow h(T-t') = s(t')$

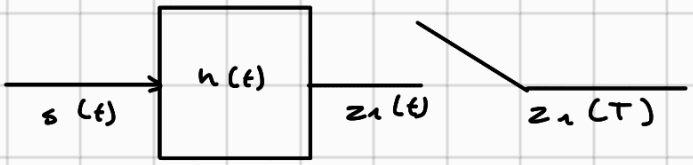
$t' = T - t$

$\Rightarrow$ $s(t) = h(T-t)$



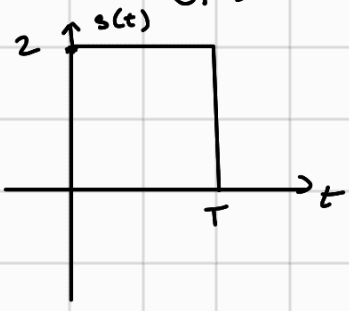


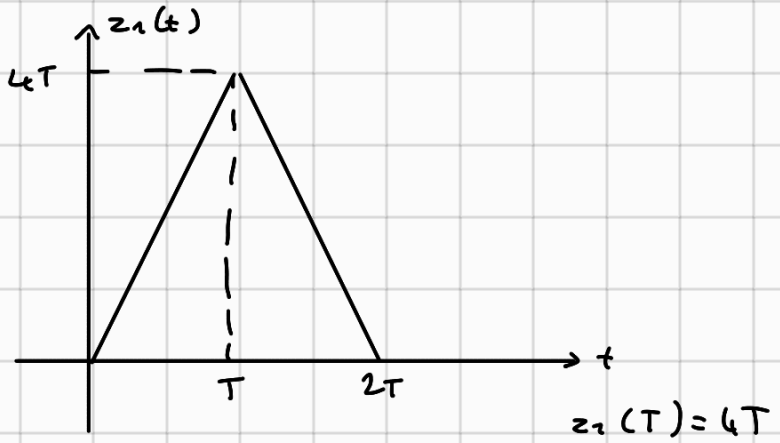
c)



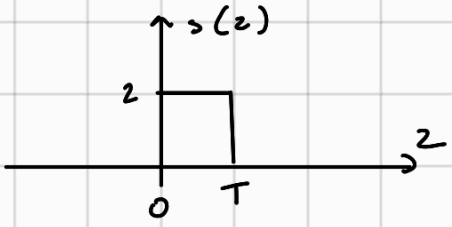
$$z_1(t) = s_1(t) \otimes h(t) = s(t) \otimes s(t)$$

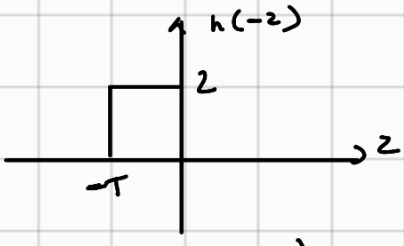
$$A\,\Pi\left(\frac{t}{T}\right) \otimes A\,\Pi\left(\frac{t}{T}\right) = A^2 T\,\Lambda\left(\frac{t}{T}\right)$$

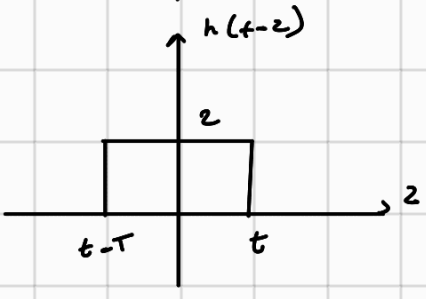




$z_1(T) = 4T$



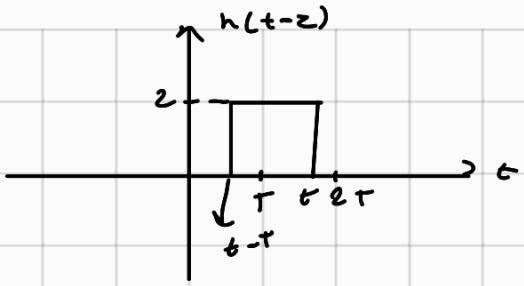




$$z_1(t) = s(t) \otimes h(t) = \int_{-\infty}^{\infty} s(z)\,h(t-z)\,dz$$

$t < 0 : z_1(t) = 0$

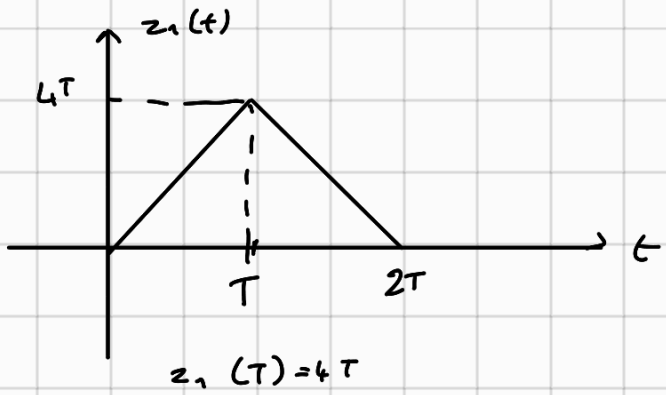
$0 < t < T : z_1(t) = \int_0^t 2 \cdot 2\,dt = 4t$



$T < t < 2T$: $z_1(t) = \int_{t-T}^{T} 2 \cdot 2\,dz =$

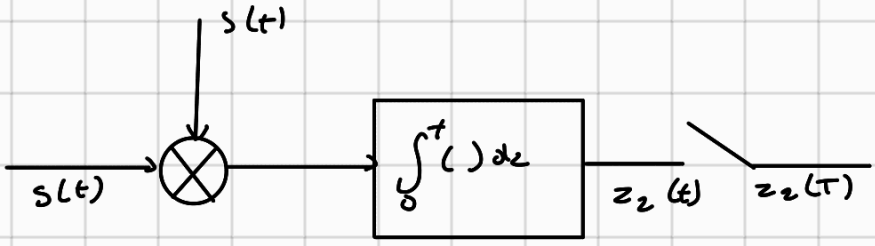
$t > 2T$: $z_1(t) = 0$

$$z_1(t) = \begin{cases} 0, & t<0 \\ 4t, & 0<t<T \\ 4(2T-t), & T<t<2T \\ 0, & t>2T \end{cases}$$



$z_1(T) = 4T$

d)

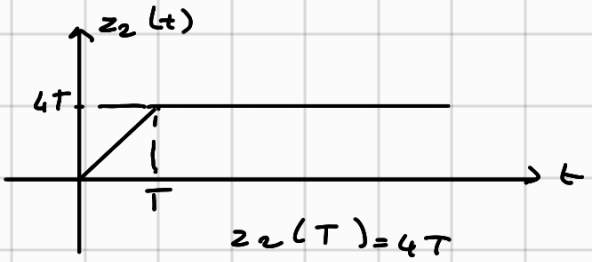


$$z_2(t) = \int_0^t s(z)\,s(z)\,dz$$

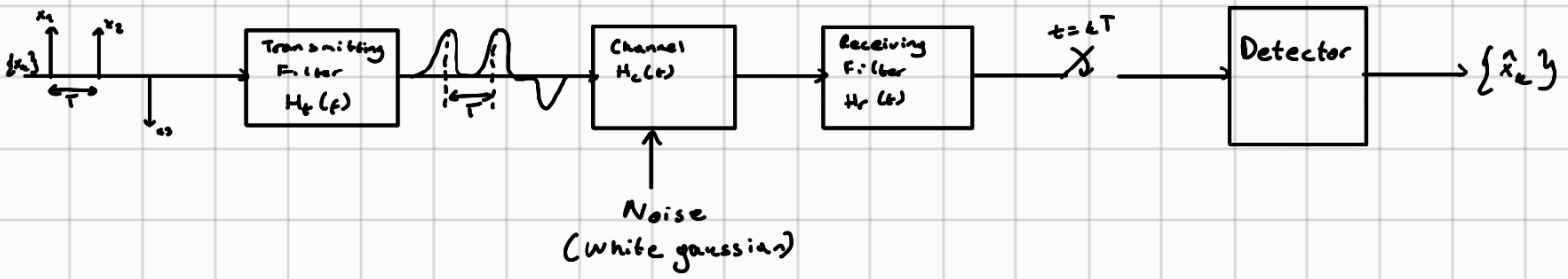
$$= \int_0^t s^2(z)\,dz = 4t \qquad t<T$$

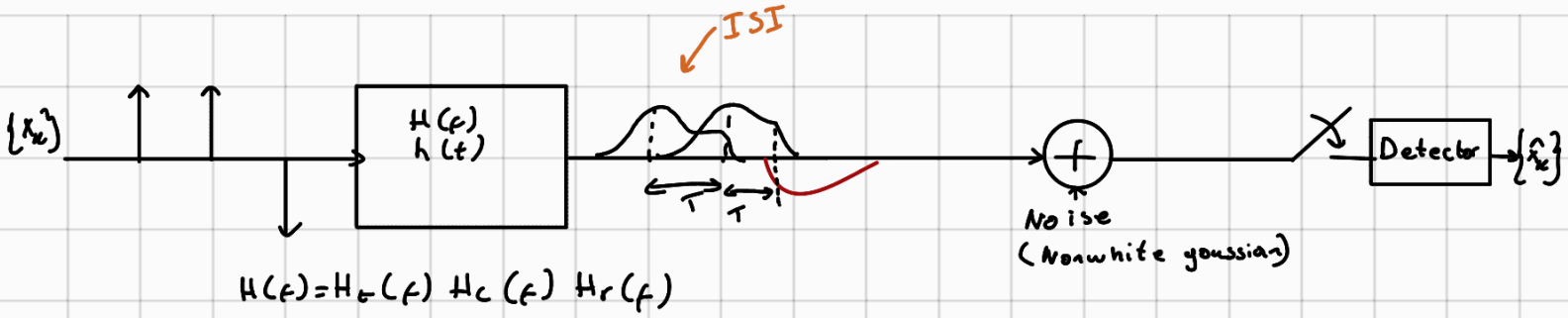
$$\int_0^T s^2(z)\,dz = 4T \qquad t>T$$

$$z_1(t) = \begin{cases} 4t & t<T \\ 4T & t>T \end{cases}$$



Semboller arası karışım (SAK)
(Intersymbol interference - ISI)
(Pulse shaping - Darbe şekillendirme)





$H(f) = H_t(f)\, H_c(f)\, H_r(f)$

Sıfır SAK için 1. Nyquist kriteri

(1st Nyquist criterion for zero ISI)

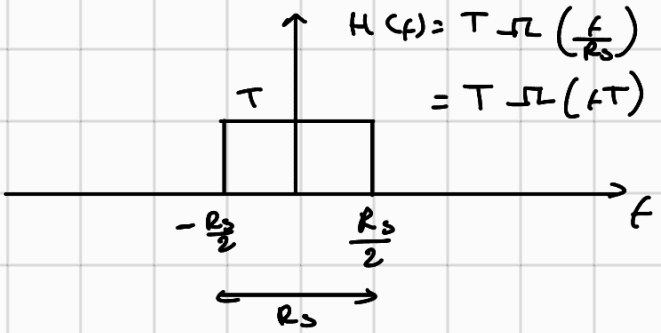
$$h(t) = \begin{cases} 1, & t = 0 \\ 0, & t = \pm nT \end{cases} \Rightarrow h(nT) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

$n = 1, 2, 3, \ldots$

$T = \frac{1}{R_s}$ ← sembol hızı

Hız: R_s $\frac{\text{sembol}}{\text{sn}}$

$\Rightarrow BG = \frac{R_s}{2}$ Hz

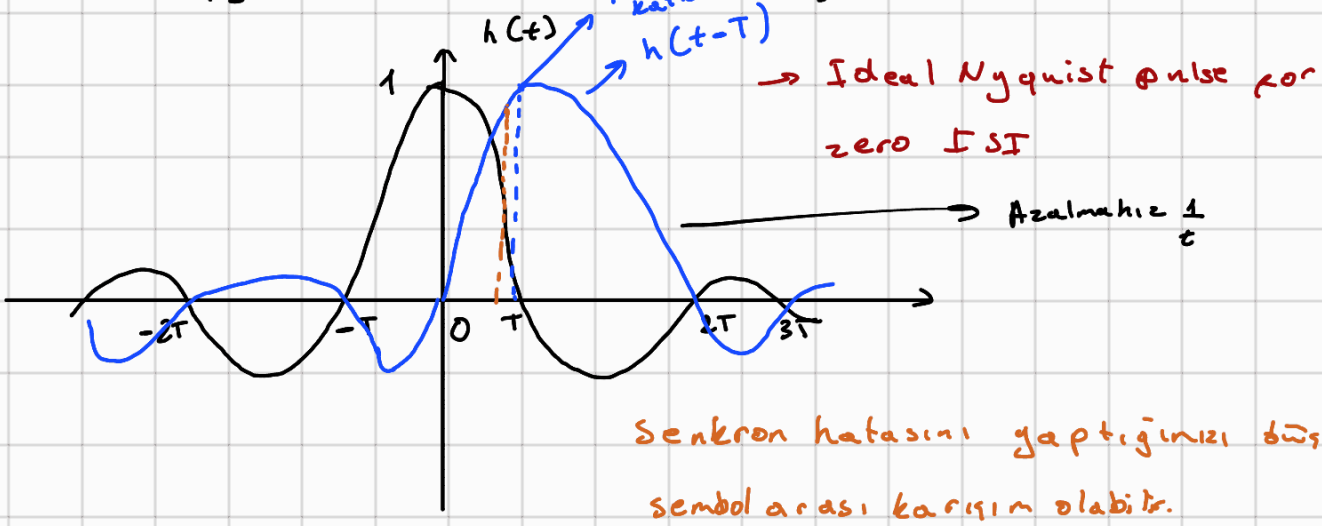


$H(f) = T\,\Pi\left(\frac{f}{R_s}\right) = T\,\Pi(fT)$

$\xrightarrow{F^{-1}}$ $h(t) = \text{sinc}\left(\frac{t}{T}\right)$

$A z\,\text{sinc}(tz) \xrightarrow{F} A\,\Pi\left(\frac{f}{z}\right)$

← ideal Nyquist filter for zero ISI



→ Ideal Nyquist pulse for zero ISI

Azalma hızı $\frac{1}{t}$

Senkron hatası yaptığınızı düşünelim sembol arası karışım olabilir.

$\frac{t}{T} = k \in \mathbb{Z} - \{0\} \Rightarrow t = kT$

Raise cosine (RC) darbesi

Daha yüksek azalma hızına sahiptir senkronizasyon hatası azalır

$BG = k\frac{R_s}{2}$ $\quad 1 \leq k \leq 2$

1. Nyquist kriterinin diğer içeriği

türetin katar.

$$h_s(t) = h(t)\, s_T(t) = h(t) \sum_k \delta(t - kT) = \sum_k h(kT)\,\delta(t - kT) = \delta(t)$$

sampled

$n(t)$

$$\mathcal{F}\left\{h(t) \sum_k \delta(t-kT)\right\} = \mathcal{F}\{\delta(t)\}$$

$$H(f) \circledast \frac{1}{T} \sum_k \delta\left(f-\frac{k}{T}\right) = 1$$

$$\frac{1}{T} \sum_k H\left(f-\frac{k}{T}\right) = 1 \Rightarrow \boxed{\sum_k H\left(f-\frac{k}{T}\right) = T}$$